

# Assignment1-111072219

111072219

2025-09-21

## PART I

### Question1

```
getwd()
```

```
## [1] "C:/Users/HSUAN/Desktop/PBA"
```

```
list.files()
```

```
## [1] "0908.R"                "0915.R"
## [3] "0915.R"                "Assignment1.log"
## [5] "Assignment1.R"          "Assignment1.Rmd"
## [7] "Assignment1.tex"        "online_retail.csv"
## [9] "PBA-Assignment-111072219.log" "PBA-Assignment-111072219.Rmd"
## [11] "PBA-Assignment-111072219.tex" "PBA.Rproj"
## [13] "Quize2.R"              "Rplot.png"
```

```
retail = read.csv("C:\\Users\\HSUAN\\Desktop\\PBA\\online_retail.csv")
head(retail)
```

| ##   | InvoiceNo | StockCode  | Description                         | Quantity | InvoiceDate  |
|------|-----------|------------|-------------------------------------|----------|--------------|
| ## 1 | 536365    | 85123A     | WHITE HANGING HEART T-LIGHT HOLDER  | 6        | 12/1/10 8:26 |
| ## 2 | 536365    | 71053      | WHITE METAL LANTERN                 | 6        | 12/1/10 8:26 |
| ## 3 | 536365    | 84406B     | CREAM CUPID HEARTS COAT HANGER      | 8        | 12/1/10 8:26 |
| ## 4 | 536365    | 84029G     | KNITTED UNION FLAG HOT WATER BOTTLE | 6        | 12/1/10 8:26 |
| ## 5 | 536365    | 84029E     | RED WOOLLY HOTTIE WHITE HEART.      | 6        | 12/1/10 8:26 |
| ## 6 | 536365    | 22752      | SET 7 BABUSHKA NESTING BOXES        | 2        | 12/1/10 8:26 |
| ##   | UnitPrice | CustomerID | Country                             |          |              |
| ## 1 | 2.55      | 17850      | United Kingdom                      |          |              |
| ## 2 | 3.39      | 17850      | United Kingdom                      |          |              |
| ## 3 | 2.75      | 17850      | United Kingdom                      |          |              |
| ## 4 | 3.39      | 17850      | United Kingdom                      |          |              |
| ## 5 | 3.39      | 17850      | United Kingdom                      |          |              |
| ## 6 | 7.65      | 17850      | United Kingdom                      |          |              |

```
str(retail)
```

```
## 'data.frame': 541909 obs. of 8 variables:
## $ InvoiceNo : chr "536365" "536365" "536365" "536365" ...
## $ StockCode : chr "85123A" "71053" "84406B" "84029G" ...
## $ Description: chr "WHITE HANGING HEART T-LIGHT HOLDER" "WHITE METAL LANTERN" "CREAM CUPID HEARTS C
## $ Quantity : int 6 6 8 6 6 2 6 6 6 32 ...
## $ InvoiceDate: chr "12/1/10 8:26" "12/1/10 8:26" "12/1/10 8:26" "12/1/10 8:26" ...
## $ UnitPrice : num 2.55 3.39 2.75 3.39 3.39 7.65 4.25 1.85 1.85 1.69 ...
## $ CustomerID : int 17850 17850 17850 17850 17850 17850 17850 17850 17850 13047 ...
## $ Country : chr "United Kingdom" "United Kingdom" "United Kingdom" "United Kingdom" ...
```

## Question 2

```
## convert InvoiceDate to Date
retail$InvoiceDate <- as.Date(retail$InvoiceDate, format = "%m/%d/%y")

## filter data
retail_filtered <- retail[retail$InvoiceDate >= as.Date('2011-07-01') &
                          retail$InvoiceDate <= as.Date('2011-08-31'),]

## number of unique invoiceNo
length(levels(as.factor(retail_filtered$InvoiceNo)))
```

```
## [1] 3664
```

## Question 3

```
## mean of quality
mean(retail_filtered$Quantity)
```

```
## [1] 10.65901
```

```
## mean of UnitPrice
mean(retail_filtered$UnitPrice)
```

```
## [1] 4.308608
```

```
## data types of each column
sapply(retail_filtered,class)
```

```
## InvoiceNo StockCode Description Quantity InvoiceDate UnitPrice
## "character" "character" "character" "integer" "Date" "numeric"
## CustomerID Country
## "integer" "character"
```

```
## number of unique values in each column
length(levels(as.factor(retail_filtered$InvoiceNo)))
```

```
## [1] 3664
```

```
length(levels(as.factor(retail_filtered$StockCode)))
```

```
## [1] 2982
```

```
length(levels(as.factor(retail_filtered$Description)))
```

```
## [1] 2953
```

```
length(levels(as.factor(retail_filtered$Quantity)))
```

```
## [1] 287
```

```
length(levels(as.factor(retail_filtered$InvoiceDate)))
```

```
## [1] 52
```

```
length(levels(as.factor(retail_filtered$UnitPrice)))
```

```
## [1] 447
```

```
length(levels(as.factor(retail_filtered$CustomerID)))
```

```
## [1] 1540
```

```
length(levels(as.factor(retail_filtered$Country)))
```

```
## [1] 28
```

## Question 4

```
## subset data of UK, Netherlands and Australia
uk <- retail_filtered[retail_filtered$Country == "United Kingdom", ]
netherlands <- retail_filtered[retail_filtered$Country == "Netherlands", ]
australia <- retail_filtered[retail_filtered$Country == "Australia", ]
```

```
## average and standard deviation of UnitPrice for uk
mean(uk$UnitPrice)
```

```
## [1] 4.359879
```

```
sd(uk$UnitPrice)
```

```
## [1] 99.41229
```

```
## average and standard deviation of UnitPrice for netherlands  
mean(netherlands$UnitPrice)
```

```
## [1] 2.862109
```

```
sd(netherlands$UnitPrice)
```

```
## [1] 12.20678
```

```
## average and standard deviation of UnitPrice for australia  
mean(australia$UnitPrice)
```

```
## [1] 2.37929
```

```
sd(australia$UnitPrice)
```

```
## [1] 2.032917
```

```
## uk_invoices  
length(levels(as.factor(uk$InvoiceNo)))
```

```
## [1] 3310
```

```
## uk_customers  
length(levels(as.factor(uk$CustomerID)))
```

```
## [1] 1375
```

```
## netherlands_invoices  
length(levels(as.factor(netherlands$InvoiceNo)))
```

```
## [1] 11
```

```
## netherlands_customers  
length(levels(as.factor(netherlands$CustomerID)))
```

```
## [1] 1
```

```
## australia_invoices  
length(levels(as.factor(australia$InvoiceNo)))
```

```
## [1] 11
```

```
## australia_customers
length(levels(as.factor(australia$CustomerID)))
```

```
## [1] 3
```

## Question 5

```
## refund customer
refund <- retail_filtered[substr(retail_filtered$InvoiceNo, start = 1, stop = 1) == "C", ]
refund <- refund[!is.na(refund$CustomerID), ]
cust_refund <- levels(as.factor(refund$CustomerID))
## number of refund customer
length(cust_refund)
```

```
## [1] 381
```

## Question 6

```
## mutate sales
retail_filtered$Sales <- retail_filtered$Quantity * retail_filtered$UnitPrice

## total sale for transactions where CustomerID is missing
na_id <- retail_filtered[is.na(retail_filtered$CustomerID), ]
head(na_id)
```

```
##      InvoiceNo StockCode      Description Quantity InvoiceDate
## 246043    558663    23574                100 2011-07-01
## 246075    558680    20711      JUMBO BAG TOYS         1 2011-07-01
## 246076    558680    21116      OWL DOORSTOP          1 2011-07-01
## 246077    558680    21166  COOK WITH WINE METAL SIGN     1 2011-07-01
## 246078    558680    21175  GIN + TONIC DIET METAL SIGN     1 2011-07-01
## 246079    558680    21206 STRAWBERRY HONEYCOMB GARLAND     1 2011-07-01
##      UnitPrice CustomerID      Country Sales
## 246043      0.00      NA United Kingdom 0.00
## 246075      4.13      NA United Kingdom 4.13
## 246076      8.29      NA United Kingdom 8.29
## 246077      4.13      NA United Kingdom 4.13
## 246078      4.96      NA United Kingdom 4.96
## 246079      3.29      NA United Kingdom 3.29
```

```
sum(na_id$Sales, na.rm = TRUE)
```

```
## [1] 173374.1
```

## PART II

### Question 7

```
## basic information of gapminder
glimpse(gapminder)

## Rows: 1,704
## Columns: 6
## $ country   <fct> "Afghanistan", "Afghanistan", "Afghanistan", "Afghanistan", ~
## $ continent <fct> Asia, Asia, Asia, Asia, Asia, Asia, Asia, Asia, Asia, ~
## $ year      <int> 1952, 1957, 1962, 1967, 1972, 1977, 1982, 1987, 1992, ~
## $ lifeExp   <dbl> 28.801, 30.332, 31.997, 34.020, 36.088, 38.438, 39.854, 40.8~
## $ pop       <int> 8425333, 9240934, 10267083, 11537966, 13079460, 14880372, 12~
## $ gdpPercap <dbl> 779.4453, 820.8530, 853.1007, 836.1971, 739.9811, 786.1134, ~

dim(gapminder)

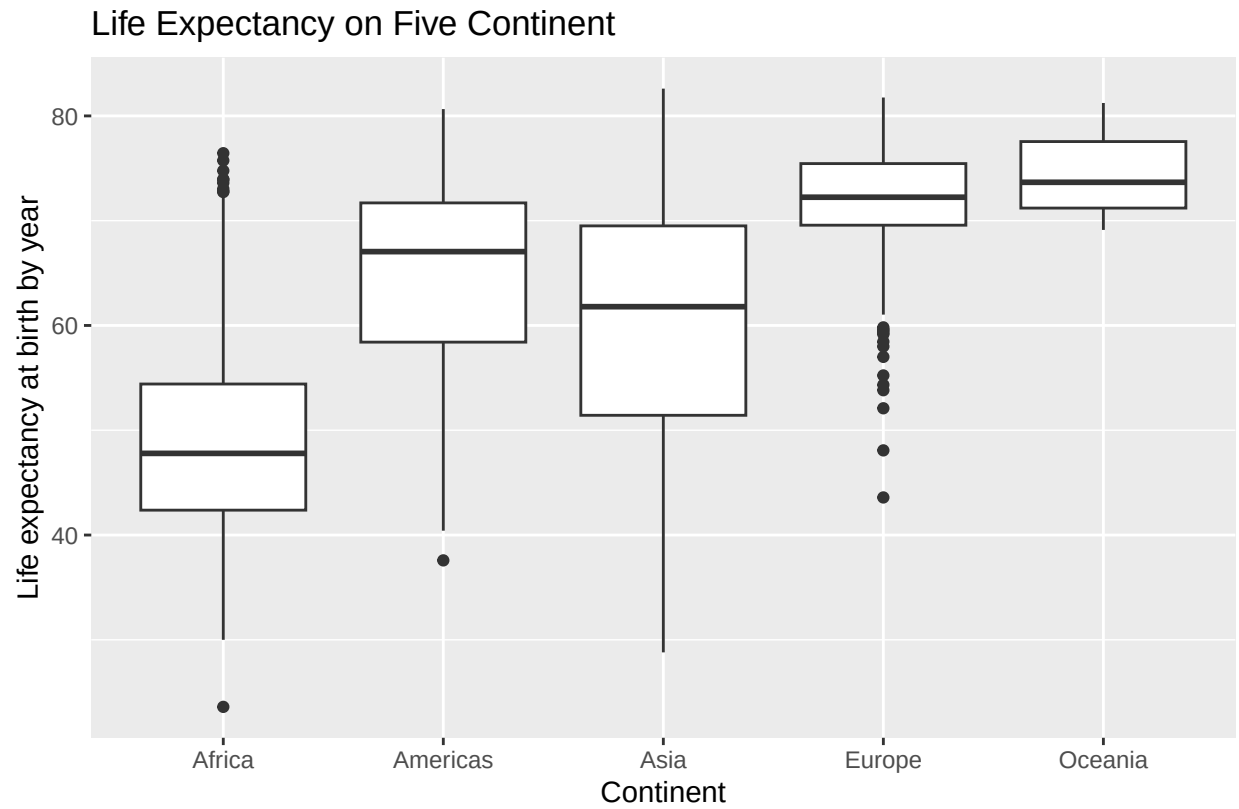
## [1] 1704    6

sapply(gapminder, class)

##   country continent   year  lifeExp    pop gdpPercap
## "factor"  "factor" "integer" "numeric" "integer" "numeric"
```

### Question 8

```
p <- ggplot(data = gapminder, aes(x = continent, y = lifeExp)) +
  geom_boxplot() +
  labs(
    title = "Life Expectancy on Five Continent",
    x = "Continent",
    y = "Life expectancy at birth by year",
    caption = "Source: Gapminder")
p
```



Source: Gapminder

### Question 9

Oceania has the highest median life expectancy. We can tell this from the horizontal line (the median line) inside the box. The median line of Oceania is at the highest.